Graph Theory | Lecture 15

Cutsets • Max-Flow–Min-Cut Theorem • Residual Graphs

From bottlenecks to maximum-flow algorithms

Learning outcomes

Define an s–t cut and calculate its capacity.

Use a cut to establish an upper bound on maximum flow.

Explain residual capacities, reverse edges, and augmenting paths.

Compare Ford–Fulkerson and Edmonds–Karp running times.

Flow network recap

Directed graph $G=(V,E)$, source s, sink t, capacity $c(u,v) \geq 0$.

Feasible flow: $0 \leq f(u,v) \leq c(u,v)$.

Flow conservation at every vertex except s and t.

Flow value $|f|$ = total flow leaving s (net) = total entering t (net).

What is an s–t cut?

A partition (S,T) of vertices with $s \in S$ and $t \in T$.

$S \cap T = \emptyset$ and $S \cup T = V$.

Cut capacity $c(S,T) = \Sigma\, c(u,v)$ over directed edges $u \in S$, $v \in T$.

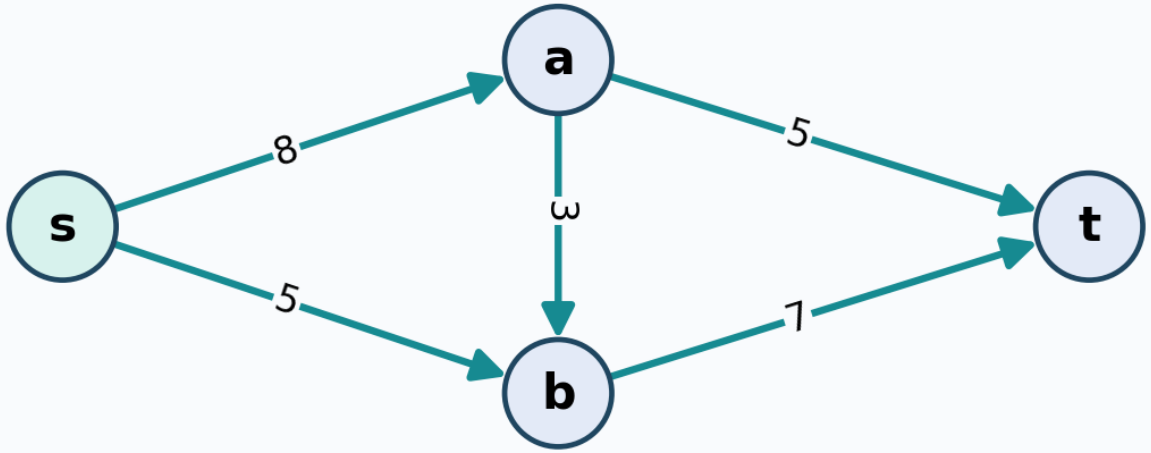
Only edges directed from S to T contribute to cut capacity.

Cutsets: the bottleneck idea

Cut separates s from t.

Only edges from S to T count.

Every feasible flow is bounded by every cut.



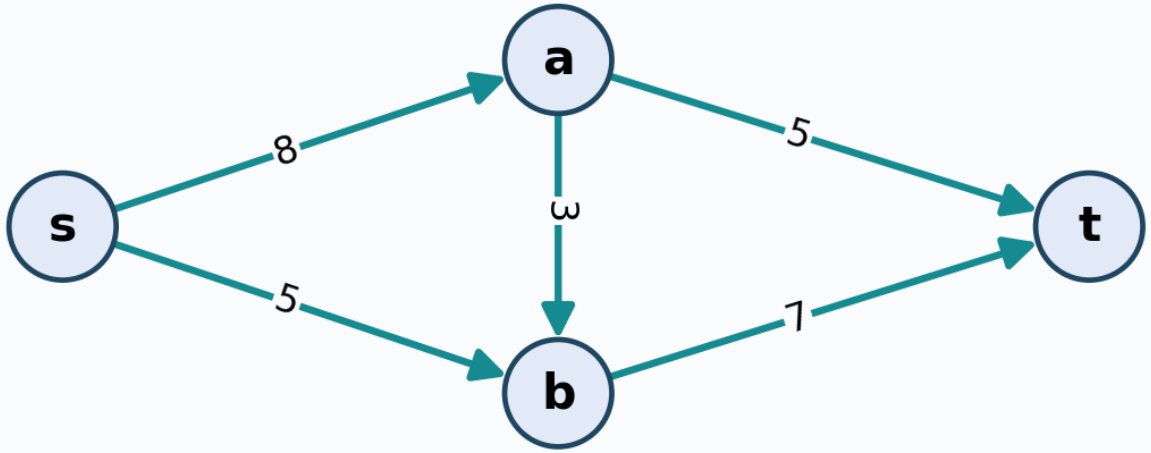
Green nodes: S side of cut

Worked network: capacities

Each label is an edge capacity.

Total capacity entering t: 5 + 7 = 12.

Can we achieve flow 12?

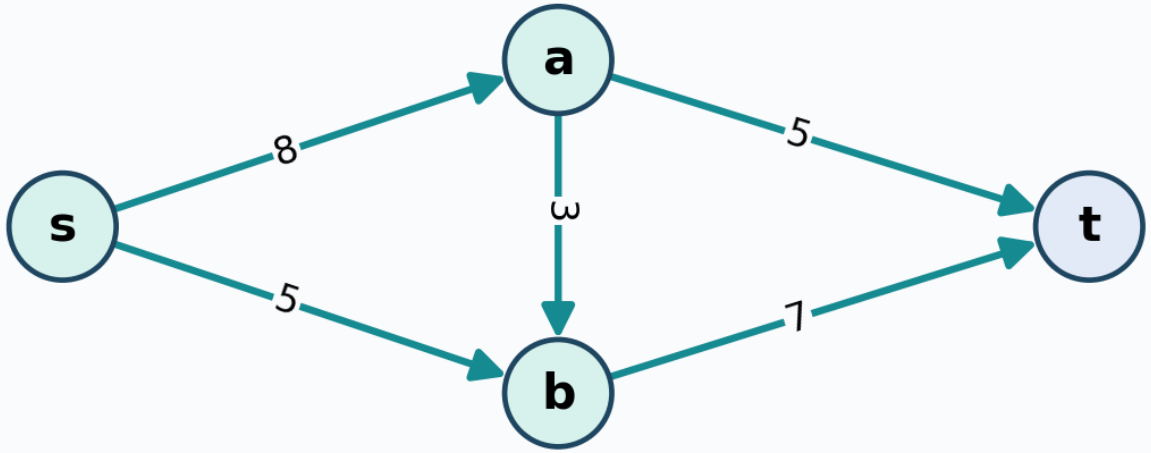


Calculate two cuts

Cut S={s}: capacity 8+5=13.

Cut S={s,a,b}: capacity 5+7=12.

The second cut gives upper bound 12.



Green nodes: S side of cut

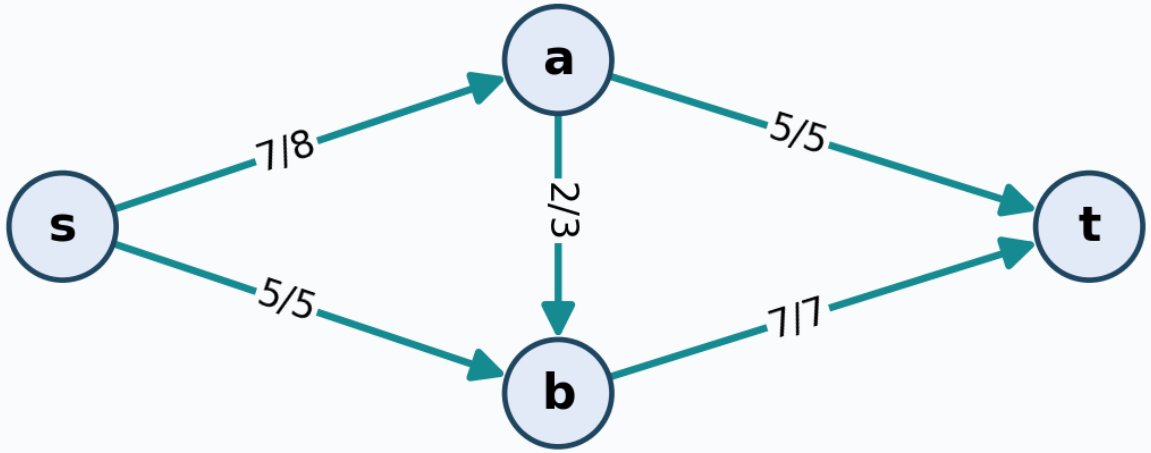
Construct a feasible flow of value 12

Edge labels: flow / capacity.

Conservation at a: 7 = 2+5.

Conservation at b: 5+2 = 7.

Total flow = 12.

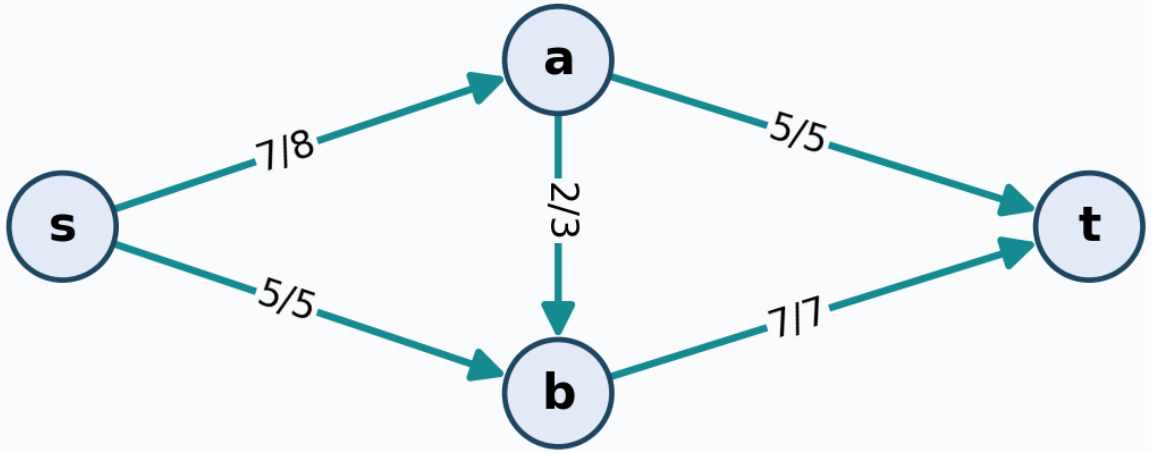


Max-Flow–Min-Cut Theorem

A feasible flow of 12 exists.

A cut of capacity 12 exists.

Therefore max flow = min cut = 12.



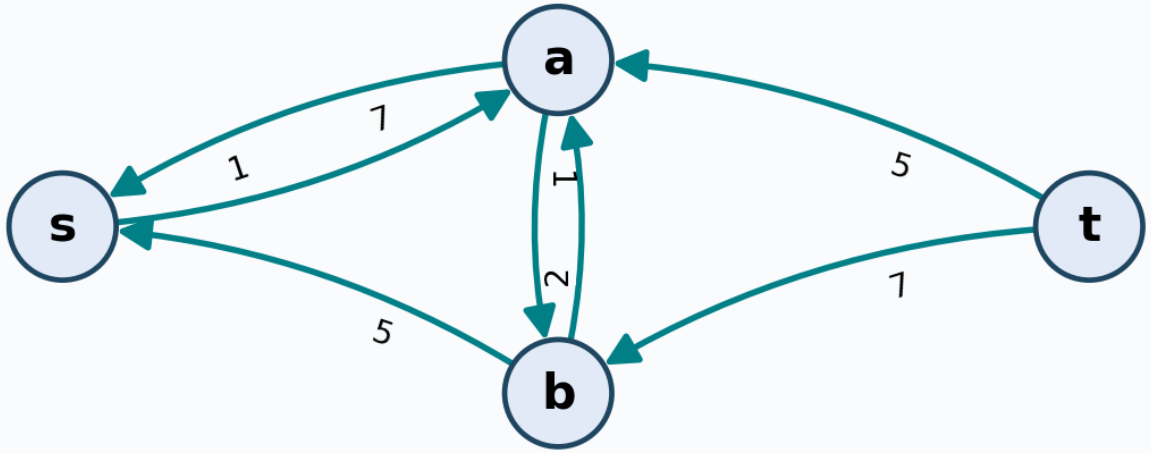
Residual graph: what changes?

Residual graph of the flow shown earlier.

Forward edge: remaining capacity.

Reverse edge: flow that can be undone.

Only positive residual capacities shown.

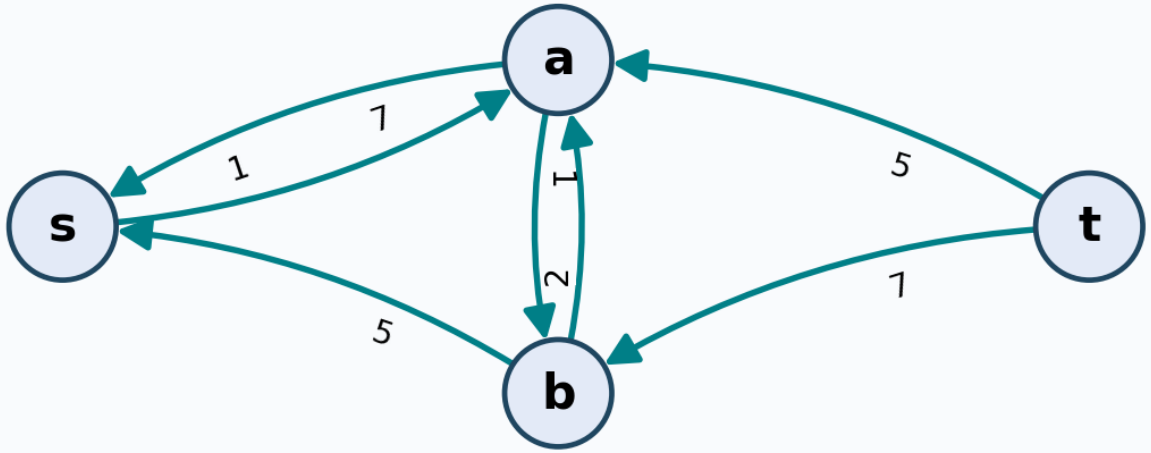


Residual example

Original a→b: capacity 3, flow 2.

Forward residual a→b: 1.

Reverse residual b→a: 2.

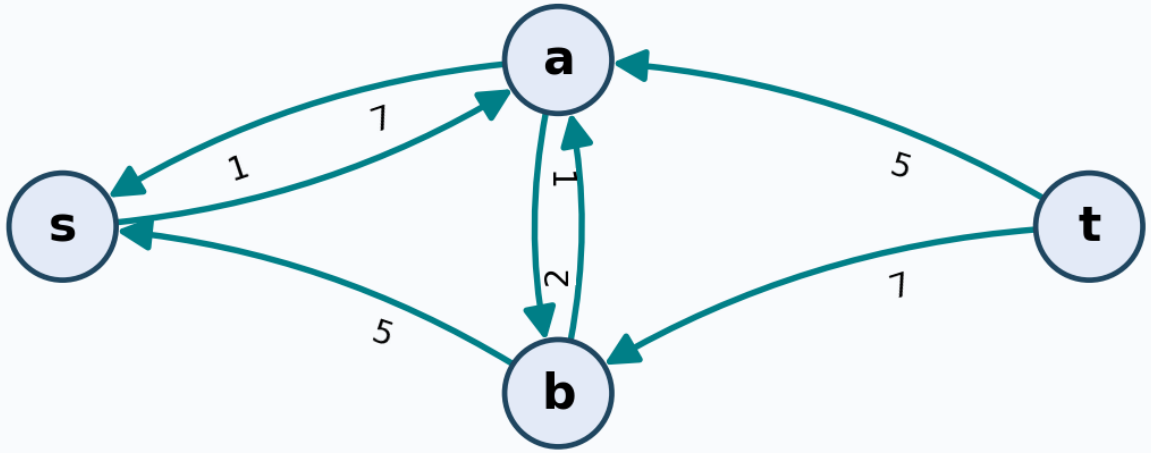


Augmenting paths

Find an s–t path in the residual graph.

Augment by the smallest residual edge capacity.

Reverse edges allow rerouting.

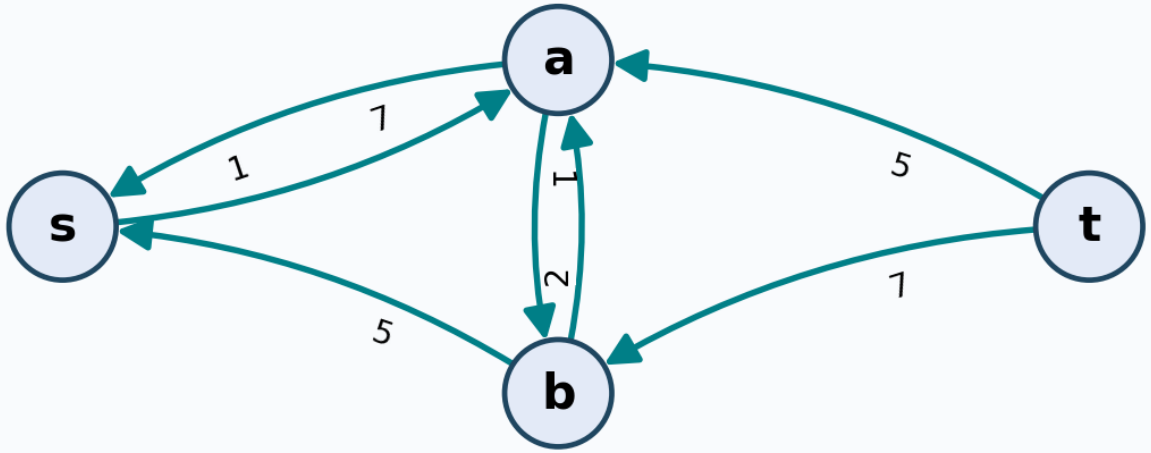


Why no augmenting path means maximum

If t is unreachable in residual graph, stop.

Reachable vertices form the S side of a cut.

Flow equals cut capacity: optimality certified.



Ford–Fulkerson: complexity

Each augmenting-path search takes O(E) time (with adjacency lists).

With integer capacities, each augmentation increases flow by ≥1.

Number of augmentations ≤ $|f^*|$ when starting from zero.

Total time $O(E \cdot |f^*|)$: pseudo-polynomial, not polynomial in input bit length.

With irrational capacities, arbitrary path choices may fail to terminate.

Edmonds–Karp: complexity

Choose each augmenting path using BFS (fewest residual edges).

Each BFS costs $O(V+E)$, commonly written $O(E)$ for relevant connected networks.

Number of augmentations is $O(VE)$.

Total time $O(VE^2)$, independent of capacity magnitudes.

This is a polynomial-time maximum-flow algorithm.

Class check (5 minutes)

Network: s→a 6, s→b 4, a→b 2, a→t 4, b→t 5.

1. Find the capacity of cut S={s,a,b}, T={t}.

2. Give an upper bound on maximum flow.

3. Can flow value 9 be achieved? Show the edge flows.

Class check: solution

Cut capacity = 4+5 = 9, so maximum flow ≤ 9.

Feasible flow: s→a 5; s→b 4; a→b 1; a→t 4; b→t 5.

Conservation: a: 5=1+4; b: 4+1=5.

Thus maximum flow = minimum cut = 9.

Lecture summary

Cutsets provide upper bounds on possible flow.

A matching feasible flow and cut capacity certify optimality.

Residual graphs capture unused capacity and flow cancellation.

Ford–Fulkerson depends on augmenting-path choices; Edmonds–Karp uses BFS.

Next: network science and graph analytics.